

HOMI BHABHA BALVAIDNYANIK COMPETITION

Class: 6 • Subject: Science • Topic: Classification of Plants

1. Why Do We Classify Plants?

There are thousands of different plants around us. They differ in size, shape, habitat, type of stem, type of root, arrangement of leaves, method of reproduction, and whether they produce flowers or seeds.

To study plants easily, scientists group plants according to their similarities and differences. This grouping is called classification.

PART A — CLASSIFICATION BASED ON HABIT

Plants can be classified according to their habit mainly into herbs, shrubs, trees, climbers and creepers.

2. Herbs

Herbs are small plants with soft, green and usually thin stems.

- Usually short in height.
- Stem is generally soft and green.
- Many herbs have medicinal or culinary uses.
- The important feature is the soft, non-woody stem—not simply the plant's height.

Examples: Mint, coriander, grass, spinach, wheat, rice, tomato and basil.

3. Shrubs

Shrubs are medium-sized, bushy plants with hard, woody stems that branch near the base.

- Generally smaller than trees.
- Have woody stems.
- Branches usually arise near the ground.
- Usually have several stems rather than one prominent main trunk.

Examples: Rose, hibiscus, cotton, lemon and henna.

Remember: Shrub → woody + bushy + branches near the ground.

4. Trees

Trees are generally tall, large plants having a strong, woody stem called a trunk.

- Usually tall.
- Have a strong, thick, woody stem.
- Usually have one main trunk.
- Branches generally arise above the ground.
- Many trees live for many years.

Examples: Mango, neem, banyan, peepal and teak.

Important: Height alone does not decide whether a plant is a tree. Stem structure and growth habit are also important.

5. Climbers

Climbers are plants with weak stems that cannot stand upright on their own and need support to grow upward.

- May climb using tendrils, twining stems, hooks or other supporting structures.
- Grow upward using external support.

Examples: Pea, grapevine, bean and money plant.

6. Creepers

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Creepers are plants with weak stems that spread along the ground.

- Weak stems.
- Cannot stand upright.
- Grow horizontally along the ground.
- Often produce fruits close to or on the ground.

Examples: Pumpkin, watermelon, muskmelon, cucumber and sweet potato.

Climbers vs Creepers: Both have weak stems. A climber uses support and grows upward, while a creeper spreads along the ground.

PART B — ROOTS

7. What is a Root?

A root is generally the underground part of a plant. It fixes the plant firmly in the soil, absorbs water and minerals, and in some plants stores food.

The two major root systems are tap root and fibrous root.

8. Tap Root

A tap root system has one main, thick root from which smaller lateral roots arise.

Structure: Main root → lateral roots → smaller roots

Examples: Carrot, radish, mustard, pea, mango and neem.

Special example: Carrot is a modified tap root that stores food.

9. Fibrous Root

A fibrous root system consists of many thin roots of approximately similar size arising from the base of the stem. There is no single prominent main root.

Examples: Grass, wheat, rice, maize and sugarcane.

Tap Root	Fibrous Root
One main root is prominent.	Many thin roots are present.
Carrot, mango, mustard	Wheat, rice, grass, maize

PART C — STEM TYPES

The stem supports leaves, flowers and fruits, carries water and minerals upward, carries prepared food to different parts, and may sometimes store food.

10. Herbaceous and Woody Stems

Herbaceous stem: Soft, green and usually non-woody. Examples: mint and coriander.

Woody stem: Hard and strong, containing wood. Examples: mango, neem and rose.

Special Modified Stems

- Potato: A modified underground stem. Its 'eyes' are buds that can grow into new shoots.
- Ginger: A rhizome, which is a modified underground stem.
- Onion: A bulb, a modified underground stem associated with fleshy storage leaves.

Homi Bhabha trap: Potato is a modified stem, not a root.

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PART D — MAJOR GROUPS OF PLANTS

For basic school-level classification, plants are grouped into Thallophyta, Bryophyta, Pteridophyta, Gymnosperms and Angiosperms.

Sequence: Thallophyta → Bryophyta → Pteridophyta → Gymnosperms → Angiosperms

11. Thallophyta

Thallophyta includes simple plant-like organisms whose body is not differentiated into true roots, stems and leaves. The body is called a thallus.

- Simple body structure.
- No true roots, stems or leaves.
- Many are aquatic or found in moist places.
- Many reproduce by spores or other simple methods.

Examples: Algae, Spirogyra and Ulothrix.

Key point: Algae do not have true roots, stems and leaves.

12. Bryophyta

Bryophytes are small, simple plants that generally grow in moist and shady places.

They are called the “amphibians of the plant kingdom” because they generally live on land but require water for reproduction.

- Small plants.
- Prefer moist habitats.
- Do not have fully developed true roots, stems and leaves.
- Reproduce by spores.

Examples: Moss, Marchantia and Funaria.

13. Pteridophyta

Pteridophytes have true roots, stems and leaves and vascular tissues, but they do not produce flowers or seeds. They reproduce mainly through spores.

- True roots, stems and leaves present.
- Vascular tissues present.
- No flowers or seeds.
- Reproduce by spores.

Examples: Fern, horsetail and club moss.

Comparison: Bryophytes are simpler; pteridophytes have true roots, stems and leaves with vascular tissue.

14. Gymnosperms

Gymnosperms are seed-producing plants whose seeds are not enclosed inside fruits. The word gymnosperm means “naked seed”.

- Produce seeds.
- Seeds are not enclosed in fruits.
- Do not produce true flowers.
- Usually woody; many are evergreen.
- Many have needle-like leaves.

Examples: Pine, Cycas, deodar and fir.

Key point: Gymnosperm → naked seeds.

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15. Angiosperms

Angiosperms are flowering plants whose seeds are enclosed inside fruits. They are the most diverse group of plants.

- Produce flowers.
- Produce seeds.
- Seeds are enclosed inside fruits.
- Have well-developed roots, stems and leaves.
- Include herbs, shrubs and trees.

Examples: Mango, rose, wheat, rice, pea, sunflower and maize.

Key point: Angiosperm → flower + fruit + enclosed seeds.

Group	True roots/stems/leaves	Flowers	Seeds	Main reproduction	Example
Thallophyta	No	No	No	Spores / simple methods	Algae
Bryophyta	Not fully developed	No	No	Spores	Moss
Pteridophyta	Yes	No	No	Spores	Fern
Gymnosperms	Yes	No true flowers	Yes, naked	Seeds	Pine
Angiosperms	Yes	Yes	Yes, enclosed	Seeds	Mango

PART E — MONOCOTS AND DICOTS

Angiosperms can be broadly divided into monocots and dicots. The main difference is the number of cotyledons in the seed.

16. Monocots

Monocotyledonous plants have one cotyledon in their seed.

- One cotyledon.
- Usually parallel leaf venation.
- Usually fibrous roots.
- Flower parts commonly occur in multiples of 3.
- Vascular bundles in the stem are generally scattered.

Examples: Rice, wheat, maize, grass, banana, coconut, onion and sugarcane.

17. Dicots

Dicotyledonous plants have two cotyledons in their seed.

- Two cotyledons.
- Usually reticulate or net-like leaf venation.
- Usually tap root.
- Flower parts commonly occur in multiples of 4 or 5.
- Vascular bundles are generally arranged in a ring.

Examples: Pea, gram, bean, mango, rose, sunflower, mustard and tomato.

Feature	Monocot	Dicot
Cotyledons	1	2

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Feature	Monocot	Dicot
Root	Usually fibrous	Usually tap root
Leaf venation	Usually parallel	Usually reticulate
Floral parts	Usually multiples of 3	Usually multiples of 4 or 5
Stem vascular bundles	Scattered	Usually in a ring
Examples	Rice, maize, wheat	Pea, mango, rose

Monocot → 1 cotyledon + parallel venation + fibrous roots

Dicot → 2 cotyledons + reticulate venation + tap root